

Suiyang Guo

+86 158-9334-6463 | 1327217178@qq.com | Northwestern Polytechnical Univ. | MSc,
Control Eng. | 2024.09—2027.06



Control Engineering MSc candidate at Northwestern Polytechnical University (Class of 2027), focused on dual-arm deformable-object manipulation. Hands-on experience across the VLA lifecycle, from data curation, SFT/offline post-training and evaluation audits to ROS2 real-robot deployment. Optimized π 0.5, X-VLA and FastWAM systems, reducing step latency from 256.5 ms to 32.05 ms ($8\times$) and idle end-effector jitter by 81%. Skilled at isolating model, data and robot-stack failures through cross-date evaluation, visual ablations and deployment safety gates.

Education

Northwestern Polytechnical University (NPU)	2024.09—2027.06
MSc, Control Engineering Robotics & AI GPA: 89.56/100	
Zhengzhou University	2020.09—2024.06
BEng, Automation (Elite Class) Robotics & AI GPA: 3.54/4.0	

Technical Skills

- **VLA / Robot Learning:** π 0.5, X-VLA, flow matching, action chunking, SFT, imitation learning, AWBC / RECAP, dataset curation, cross-domain evaluation, FastWAM world-action models
- **Model & Inference Systems:** PyTorch (DDP), JAX/Flax (FSDP), DINOv3 / SigLIP; Triton kernels, CUDA Graphs, torch.compile, RTC and end-to-end latency profiling
- **Programming:** C/C++ (proficient), Python (proficient), MATLAB (fluent); Linux driver/application development
- **Robotics:** ROS/ROS2, dual-arm manipulation and bring-up, IK, gripper/action mapping, MuJoCo, navigation, SLAM, motion planning and point clouds
- **Embedded & Hardware:** Yocto / i.MX6ULL embedded Linux, STM32F4, FreeRTOS, motor control and PCB design
- **Engineering:** Git, Linux, WebSocket, PyQt5 host applications and Autodesk Inventor

Experience

IDEA Research: Visincept	2026.04—Present
VLA & World Model Intern, Embodied AI Team	
• Built a dual-Piper cloth-manipulation pipeline on openpi/π0.5 spanning data curation, SFT, offline evaluation and ROS2 deployment; reduced native-validation MAE@50 from 0.0337 to 0.0079 and exposed a $3.2\times$ cross-date generalization degradation through a separate audit split • Led VLA inference/control optimization using RTC chunking, torch.compile, CUDA Graphs and Triton autotuning; reduced step latency from 256.5 ms to 32.05 ms ($8\times$), while minimum-jerk/EMA smoothing cut idle end-effector jitter from 3.33 mm to 0.63 mm (-81%) • Brought up X-VLA on the dual-arm platform; designed visual-ablation gates that traced a silent video-path fallback to black frames, raising the vision-to-proprioception sensitivity ratio from 0 to 5.17 after the fix; firmware-level IK reduced the control cycle from 1.4 s to 0.33 s • Implemented an AWBC/RECAP-advantage offline post-training pipeline and adapted RECAP/RLT to a robotic nail-painting task; reproduced FastWAM with WebSocket/ROS2, continuous-gripper support and preflight safety gates; built CRAVE milestone/value signals with 0.865 stage-progress correlation	
Zhongqin Herui (Shaanxi) Technology Co., Ltd.	2025.11—2026.01
Navigation & Mapping Algorithm Engineer (Part-time)	
• Developed 3D mapping and navigation for an orchard autonomous inspection vehicle • Owned equipment/sensor selection, calibration, and sensor-data processing • Built 3D maps from raw LiDAR, then occupancy grids for vehicle navigation	
Huawei Digital Energy Bootcamp	2025.07
Trainee (Training)	
• Studied digital-energy cloud microservice (DDD) architecture and AI applications/engineering in energy services • Practiced prompt engineering and DevOps deployment workflows	

Projects

- Series-Parallel Hybrid Wheeled Humanoid Robot

2024.09—Present

National Key Research Project | Key Provincial Lab —team member

- Built a wheeled humanoid comprising an AGV base, parallel mechanism, dual 6-DoF arms and a 2-DoF head; created a MuJoCo simulation/control-integration environment and a ROS master controller for mapping, planning, obstacle avoidance, point-cloud processing and IK
 - Developed in-house dual-arm and chassis motor-control systems and integrated STM32F4 firmware with actuators; built a Yocto/i.MX6ULL embedded-Linux image and ROS multi-machine networking, doubling the arm control frequency
- AI-LLM Voice-Interaction Robot

2023.12—2024.09

Outstanding Undergraduate Thesis | Solo developer

- Independently built a Raspberry Pi/PyQt5/ROS voice robot integrating audio capture, ASR, LLM dialogue, speech synthesis, digital-human lip sync and smart-home control; received 2 software copyrights
- TCM Therapy Device Series (3 generations)

2021.03—2023.09

"Challenge Cup" Provincial 2nd Prize | Core R&D member

- Products:** smart gourd-moxibustion (gen 1) → warm-needle therapy (gen 2) → home mobile moxibustion (gen 3)
 - Stack:** STM32, FreeRTOS, fuzzy PID, sensor fusion, Bluetooth
 - Work:** led full-stack R&D across three generations (mechanical → circuit → embedded → host software); STM32 firmware (FreeRTOS, fuzzy-PID, sensor integration); hardware circuits (Bluetooth, servo drive, smoke handling); PCB design; multi-sensor (temperature/image/pressure) integration; servo & stepper motor drivers

Awards & Honors

- "Challenge Cup" Henan Collegiate Academic Sci-Tech Works Competition (16th)

Provincial 2nd Prize (2023)
- 2nd Collegiate Electrical & Electronic Engineer Innovation Contest

Central-China 1st Prize (2023)
- 16th "Siemens Cup" Intelligent Manufacturing Challenge

Western Region 2nd Prize (2022)
- Lanqiao Cup —Embedded Design & Development

Provincial 3rd Prize (2022)
- HIT Zhengzhou Research Institute Training Camp

Outstanding Camper (2022)
- Zhengzhou Univ. 2nd-Class Scholarship & Merit Student

2022—2023, 2024—2025

Patents & Certificates

- 4 invention patents, 2 utility-model patents, 2 software copyrights
 - CET-6 (College English Test Band 6): 473

